

Saeyoung Choi

Ph.D. Candidate in Seoul National University

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EDUCATION

Seoul National University, Seoul, South Korea

Mar 2023 – Present

- Technology Management, Economics and Policy Department, College of Engineering
- Ph.D. Candidate
- Thesis: - (Advisor: Junghye Lee)

Yonsei University, Seoul, South Korea

Mar 2021 – Feb 2023

- M.S. in Mechanical Engineering
- Thesis: Thermodynamic analysis of solid oxide electrolysis cell system and its thermal integration (Advisor: Jongsup Hong)

Yonsei University, Seoul, South Korea

Mar 2015 – Feb 2021

- B.S. in Mechanical Engineering
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RESEARCH INTERESTS

- Science & Technology Policy
- Researchers' Career
- Flexibility of Innovation System

RESEARCH METHODS

- Machine Learning & Deep Learning on Sequential Data
 - Machine Learning & Deep Learning on Graph
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PEER-REVIEWD PUBLICATIONS

- **Choi, S.**, & Hong, J. (2023). Versatile thermodynamic design of a solid oxide electrolysis cell system and its sensitivity to operating conditions. *Energy Conversion and Management*, 298, 117776.
- Min, G.*, **Choi, S.***, & Hong, J. (2022). A review of solid oxide steam-electrolysis cell systems: Thermodynamics and thermal integration. *Applied Energy*, 328, 120145. (*: equally contributed)
- Min, G., Park, Y. J., **Choi, S.**, & Hong, J. (2021). Sensitivity analysis of a solid oxide co-electrolysis cell system with respect to its key operating parameters and optimization with its performance map. *Energy Conversion and Management*, 249, 114848.

WORKING PAPERS

- Revealing country-level disparities in researchers' transition to artificial intelligence: A deep learning-based analysis

PATENTS

- J. Hong, G.B. Min, **S. Choi**, J. Lee, K. Jeong. "Multidimensional analysis system and method of fuel cell stack". **KR 10-2023-0087682** (2023).
 - J. Hong, **S. Choi**, J. Lee, K. Jeong. "Sequential-modular simulation apparatus and method for fuel cell system analysis". **KR 10-2023-0077377** (2023).
 - J. Hong, G.B. Min, **S. Choi**, J. Lee, K. Jeong, Y.J. Park. "Fluid stream property evaluation apparatus and method for a fuel cell system simulation". **KR 10-2023-0077376** (2023).
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INDUSTRY & OTHER EXPERIENCE

H-Cube Solutions, Seoul, South Korea

Nov 2021 – Dec 2022

- Lab spin-off (Technology Commercialization Project)
- Developed engineering software for fuel cell and electrolysis system design and analysis (source code for sequential simulation, GUI development)
- Prototype showcased at **CES 2023**
- Delivered software to industrial partners